

Section 4J. Classification Errors

Victor M. Preciado, PhD MIT EECS
Dept of Electrical & Systems Engineering
University of Pennsylvania
preciado@seas.upenn.edu

Classification Errors

- ▶ Consider a classification problem with these ingredients:
 - ▶ Two classes $\mathcal{C} = \{0, 1\}$
 - ▶ A training dataset $\mathcal{D}_{\text{Tr}} = \{(\mathbf{x}_i, y_i)\}$ with $y_i \in \{0, 1\}$
 - ▶ A classifier $C : \mathbb{R}^p \rightarrow \{0, 1\}$
- ▶ The output of a classifier can be categorized as follows:
 - ▶ If $C(\mathbf{x}_i) = y_i = 1$, we say that point i is a *true positive* (TP)
 - ▶ If $C(\mathbf{x}_i) = y_i = 0$, we say that point i is a *true negative* (TN)
 - ▶ If $C(\mathbf{x}_i) = 1$ and $y_i = 0$, we say that point i is a *false positive* (FP)
 - ▶ If $C(\mathbf{x}_i) = 0$ and $y_i = 1$, we say that point i is a *false negative* (FN)

Classification Errors (cont.)

- ▶ The total number of errors made by your classifier is equal to the sum of FPs and FNs, which can be counted as:

$$|\text{FP}| = \sum_{i=1}^N \delta(C(\mathbf{x}_i) - 1) \delta(y_i)$$

$$|\text{FN}| = \sum_{i=1}^N \delta(C(\mathbf{x}_i)) \delta(y_i - 1)$$

where $\delta(x)$ is called a Dirac's delta function, defined as $\delta(x) = 1$ for $x = 0$ and $\delta(x) = 0$ for $x \neq 0$.

- ▶ The *classification error rate* is defined as

$$\text{Err}_{\text{Tr}} = \frac{|\text{FP}| + |\text{FN}|}{N}$$

Classification Errors: LDA Theory

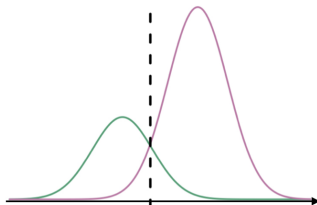
Analysis of errors in LDA with one input and two classes:

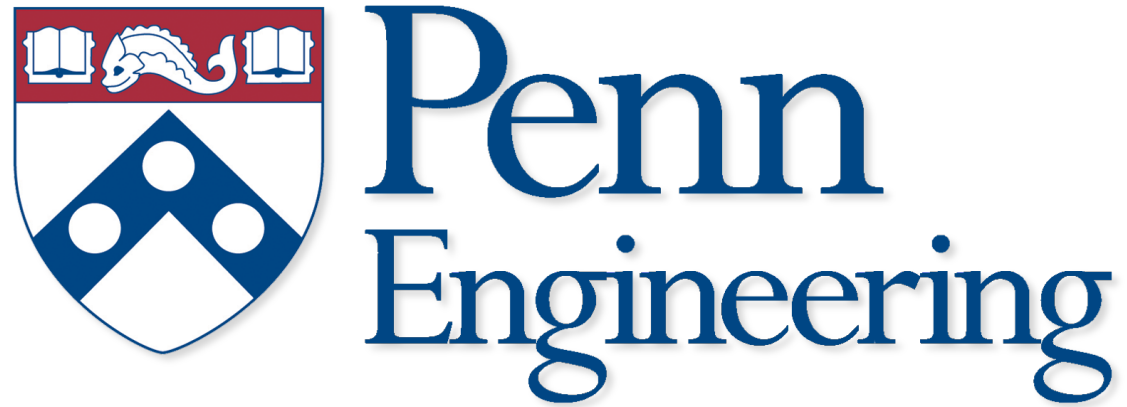
- ▶ $\Pr(Y = 0) = \pi_0$ and $\Pr(Y = 1) = \pi_1$
- ▶ $\Pr(X = x|Y = k) = f_k(x)$ with $k \in \{0, 1\}$

How much is the *expected* classification error rate of the Bayes optimal classifier $C(x)$?

Remember,

$$C(x) = \begin{cases} 0 & \text{if } \pi_0 f_0(x) > \pi_1 f_1(x) \iff x < \gamma \\ 1 & \text{if } \pi_1 f_1(x) \geq \pi_0 f_0(x) \iff x \geq \gamma \end{cases}$$





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